Solution

A1. The full energy consists of the potential energy of deformed springs and the kinetic energy of atomic motion, so

$$\boxed{E = \sum_n \left(\frac{k(u_n - v_n)^2}{2} + \frac{\alpha k(u_{n+1} - v_n)^2}{2} + \frac{\beta k(u_{n+1} - u_n)^2}{2} + \frac{\beta k(v_{n+1} - v_n)^2}{2} + \frac{m\dot{u}_n^2}{2} + \frac{m\dot{v}_n^2}{2} \right)}$$

A2. Differentiating the energy expression, we obtain

$$\dot{E} = \sum_n \Big(k(u_n - v_n)(\dot{u}_n - \dot{v}_n) + \alpha k(u_{n+1} - v_n)(\dot{u}_{n+1} - \dot{v}_n) + \beta k(u_{n+1} - u_n)(\dot{u}_{n+1} - \dot{u}_n) + \beta k(v_{n+1} - v_n)(\dot{v}_{n+1} - \dot{v}_n) + m\dot{u}_n\ddot{u}_n + m\dot{v}_n\ddot{v}_n \Big).$$

Grouping the terms containing $\dot{v}_n$ and $\dot{u}_n$

$$\dot{E} = \sum_n \Big(\dot{v}_n(m\ddot{v}_n + kv_n(\alpha + 1 + 2\beta) - ku_n - \alpha k u_{n+1} - \beta k(v_{n-1} + v_{n+1})) + \dot{u}_n(m\ddot{u}_n + ku_n(\alpha + 1 + 2\beta) - kv_n - \alpha k v_{n-1} - \beta k(u_{n-1} + u_{n+1})) \Big).$$

Here we used that

$$\sum_n \dot{v}_{n+1}(v_{n+1} - v_n) = \sum_n \dot{v}_n(v_n - v_{n-1}), \qquad \sum_n \dot{u}_{n+1}(u_{n+1} - v_n) = \sum_n \dot{u}_n(u_n - v_{n-1}),$$

and

$$\sum_n \dot{u}_{n+1}(u_{n+1} - u_n) = \sum_n \dot{u}_n(u_n - u_{n-1}).$$

Setting the coefficients A and B to zero leads to the system of differential equations.

$$\boxed{\begin{cases} m\ddot{v}_n + kv_n(\alpha + 1 + 2\beta) - ku_n - \alpha k u_{n+1} - \beta k(v_{n-1} + v_{n+1}) = 0, \\ m\ddot{u}_n + ku_n(\alpha + 1 + 2\beta) - kv_n - \alpha k v_{n-1} - \beta k(u_{n-1} + u_{n+1}) = 0. \end{cases}}$$

We recognize that the derived equations describe Newtonian dynamics of the two atoms in the unit cell.

A3. Using $u_n(t) = U \cdot \exp(-i\omega t + iqn)$, $v_n(t) = V \cdot \exp(-i\omega t + iqn)$ we obtain expressions for the second time derivatives of the displacements

$$\ddot{u}_n(t) = -\omega^2 U \cdot \exp(-i\omega t + iqn), \quad \ddot{v}_n(t) = -\omega^2 V \cdot \exp(-i\omega t + iqn).$$

Substituting these expressions into system from A2 and dividing by $\exp(-i\omega t + iqn)$. We obtain

$$\begin{cases} -\omega^2 mV + kV(\alpha + 1 + 2\beta) - kU - \alpha k U \exp(iq) - \beta k V(\exp(-iq) + \exp(iq)) = 0, \\ -\omega^2 mU + kU(\alpha + 1 + 2\beta) - kV - \alpha k V \exp(-iq) - \beta k U(\exp(-iq) + \exp(iq)) = 0. \end{cases}$$

After regrouping the terms, the system takes the required form

$$\begin{cases} k(1 + \alpha \exp(iq)) \cdot U + (\omega^2 m - k(\alpha + 1 + 2\beta) + \beta k(\exp(-iq) + \exp(iq))) \cdot V = 0, \\ (\omega^2 m - k(\alpha + 1 + 2\beta) + \beta k(\exp(-iq) + \exp(iq))) \cdot U + k(1 + \alpha \exp(iq)) \cdot V = 0. \end{cases}$$

Using Euler's formula, we find that $\exp(-iq)+\exp(iq)=2\cos q$. The system transforms to

$$\boxed{\begin{cases} k(1+\alpha\exp(iq))\cdot U+(\omega^2 m-k(\alpha+1+2\beta)+2\beta k\cos q)\cdot V=0, \\ (\omega^2 m-k(\alpha+1+2\beta)+2\beta k\cos q)\cdot U+k(1+\alpha\exp(-iq))\cdot V=0. \end{cases}}$$

A4. If a phonon can propagate in the chain, then non-trivial solutions to system $A3$ exist which implies the condition $ad-bc=0$. Therefore

$$k^2(1+\alpha\exp(iq))(1+\alpha\exp(-iq))=(\omega^2 m-k(\alpha+1+2\beta)+2\beta k\cos q)^2.$$

The direct consequence of Euler's formula is

$$(1+\alpha\exp(iq))(1+\alpha\exp(-iq))=1+\alpha^2+\alpha(\exp(iq)+\exp(-iq))=1+2\alpha\cos q+\alpha^2.$$

Substitute this into the original equation and divide by k^2, hence we obtain

$$1+2\alpha\cos q+\alpha^2=\left(\frac{\omega^2}{\omega_0^2}-(\alpha+1+2\beta)+2\beta\cos q\right)^2.$$

Therefore

$$\pm\sqrt{1+2\alpha\cos q+\alpha^2}=\frac{\omega^2}{\omega_0^2}-(\alpha+1+2\beta)+2\beta\cos q,$$

and the dispersion relation takes the form

$$\boxed{\frac{\omega}{\omega_0}=\sqrt{\pm\sqrt{1+2\alpha\cos q+\alpha^2}-2\beta\cos q+\alpha+1+2\beta}}$$

A5. For $q\to 0$ the approximation $\cos q\simeq 1-\frac{q^2}{2}$ holds. The dispersion relation takes the form

$$\frac{\omega}{\omega_0}\simeq\sqrt{\pm\sqrt{1+2\alpha-\alpha q^2+\alpha^2}+\beta q^2+\alpha+1}=\sqrt{(1+\alpha)\left(\pm\sqrt{1-\frac{\alpha q^2}{(1+\alpha)^2}}+\frac{\beta q^2}{1+\alpha}+1\right)}.$$

Expanding the square root in a Taylor series

$$\frac{\omega}{\omega_0}\simeq\sqrt{(1+\alpha)\left(\pm\left(1-\frac{\alpha q^2}{2(1+\alpha)^2}\right)+\frac{\beta q^2}{1+\alpha}+1\right)}.$$

Next, we examine the first phonon solution

$$\left(\frac{\omega}{\omega_0}\right)_1\simeq\sqrt{\frac{\alpha q^2}{2(1+\alpha)}+\beta q^2}.$$

$$\boxed{\left(\frac{\omega}{\omega_0}\right)_1\simeq|q|\sqrt{\frac{\alpha}{2(1+\alpha)}+\beta}}$$

For the second phonon

$$\frac{\omega}{\omega_0}\simeq\sqrt{2(1+\alpha)\left(1-\frac{\alpha q^2}{4(1+\alpha)^2}+\frac{\beta q^2}{2(1+\alpha)}\right)}.$$

Again, we expand the square root in a Taylor series

$$\boxed{\left(\frac{\omega}{\omega_0}\right)_2 \simeq \sqrt{2(1+\alpha)}\left(1+\frac{1}{4}\left(\frac{\beta}{1+\alpha}-\frac{\alpha}{2(1+\alpha)^2}\right)q^2\right)}$$

A6. Using the variable $p = q - \pi$. The dispersion relation takes the form

$$\frac{\omega}{\omega_0} \simeq \sqrt{\alpha+1+4\beta-\beta p^2 \pm \sqrt{1-2\alpha+\alpha p^2+\alpha^2}} = \sqrt{\alpha+1+4\beta-\beta p^2 \pm (1-\alpha)\sqrt{1+\frac{\alpha p^2}{(1-\alpha)^2}}}.$$

Therefore, for the acoustic phonon

$$\frac{\omega}{\omega_0} \simeq \sqrt{2\alpha+4\beta-\left(\beta+\frac{\alpha}{2(1-\alpha)}\right)p^2} = \sqrt{2\alpha+4\beta}\sqrt{1-\frac{1}{2(\alpha+2\beta)}\left(\beta+\frac{\alpha}{2(1-\alpha)}\right)p^2}.$$

Again, we expand the square root in a Taylor series

$$\boxed{\left(\frac{\omega}{\omega_0}\right)_1 \simeq \sqrt{2\alpha+4\beta}\left(1-\frac{1}{4(\alpha+2\beta)}\left(\beta+\frac{\alpha}{2(1-\alpha)}\right)p^2\right)}$$

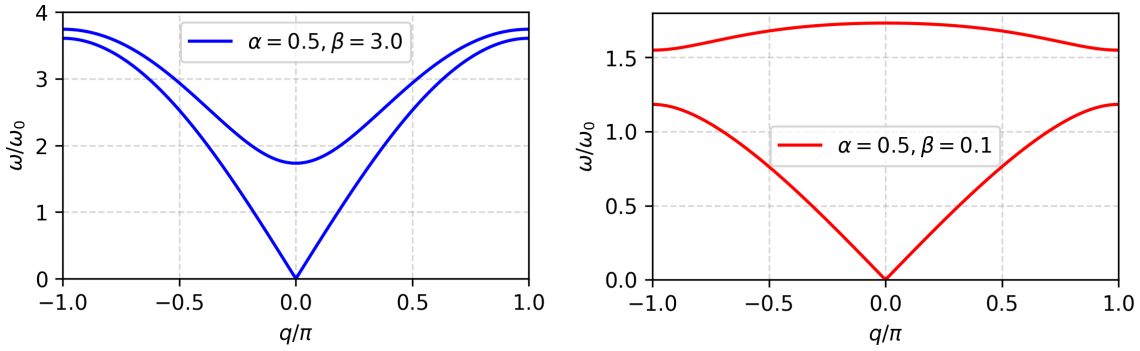
For the optical phonon:

$$\frac{\omega}{\omega_0} \simeq \sqrt{2+4\beta-\left(\beta-\frac{\alpha}{2(1-\alpha)}\right)p^2} \simeq \sqrt{2+4\beta}\sqrt{1-\frac{1}{2(1+2\beta)}\left(\beta-\frac{\alpha}{2(1-\alpha)}\right)p^2}.$$

Again, we expand the square root in a Taylor series

$$\boxed{\left(\frac{\omega}{\omega_0}\right)_2 \simeq \sqrt{2+4\beta}\left(1-\frac{1}{4(1+2\beta)}\left(\beta-\frac{\alpha}{2(1-\alpha)}\right)p^2\right)}$$

A7.



B1. Substituting the solutions in the form of plane waves, we obtain

$$u_n^\Sigma = \mu u_n + \eta \tilde{u_n} = U\exp(-i\omega t)\left(\mu\exp(iqn)+\eta\exp(-iqn)\right),$$

and similarly

$$v_n^\Sigma = \mu v_n + \eta \tilde{v_n} = V\exp(-i\omega t)\left(\mu\exp(iqn)+\eta\exp(-iqn)\right).$$

The conditions $u_0^\Sigma = 0$ and $v_0^\Sigma = 0$ turn out to be equivalent to each other, just as the conditions $u_N^\Sigma = 0$ and $v_N^\Sigma = 0$ are. Thus,

$$\boxed{\begin{cases}\mu+\eta=0,\\ \mu\exp(iqN)+\eta\exp(-iqN)=0.\end{cases}}$$

B2. To answer this question, one can use the technique described in part A of this problem. However, it is simpler to substitute $\eta = -\mu$ from the first equation into the second and divide by μ, taking into account the existence of a nontrivial solution. We obtain $\exp(iqN) = \exp(-iqN)$, which is equivalent to $\exp(2iqN) = 1$, i.e., $2qN = 2\pi l$, $l \in \mathbb{Z}$. Note that at $q = 0$ the phonon amplitude becomes zero, so $l = 0$ must be excluded from the answer.

$$\boxed{q = \frac{\pi l}{N}, \quad l \in \mathbb{Z}, l \neq 0}$$

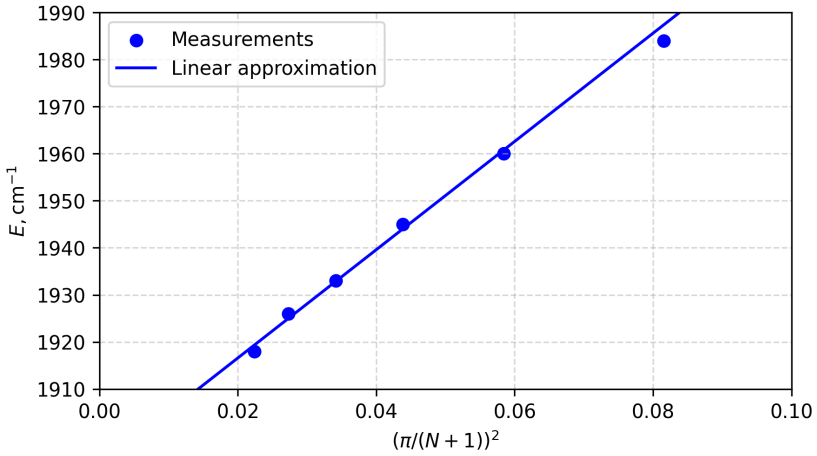
B3. From the condition, it follows that the quasimomentum q can be expressed as $q = \frac{\pi l}{N+1}$, $l \in \mathbb{Z}$ (since after applying the described technique, the chain contains $N + 2$ unit cells). The low-energy phonon corresponds to the minimum absolute value of quasimomentum, i.e., $q = \frac{\pi}{N+1}$.

Using the provided experimental plot, we extract the dependence of the low-energy phonon's energy on N. We employ the Taylor series expansion of ω/ω_0 near the point $q = 0$. Thus,

$$\frac{\omega}{\omega_0} \simeq \sqrt{2(1+\alpha)}\left(1 + \left(\frac{\beta}{4(1+\alpha)} - \frac{\alpha}{8(\alpha+1)^2}\right)\left(\frac{\pi}{N+1}\right)^2\right).$$

N	E, cm^{-1}	$\pi^2/(N+1)^2$
20	1918	0.0224
18	1926	0.0273
16	1933	0.0341
14	1945	0.0439
12	1960	0.0584
10	1984	0.0816

Let's plot the dependence $E\,(x = (\pi/(N+1))^2)$. As theoretically expected, it turned out to be linear. From the plot, we find the equation of the approximating line $E = (1150x + 1900)\,(\mathrm{cm}^{-1})$.



From the above, we obtain the system of equations

$$\begin{cases} 1150\ \mathrm{cm}^{-1} = \hbar\omega_0\sqrt{2(1+\alpha)}\left(\dfrac{\beta}{4(1+\alpha)} - \dfrac{\alpha}{8(\alpha+1)^2}\right), \\ 1900\ \mathrm{cm}^{-1} = \hbar\omega_0\sqrt{2(1+\alpha)}. \end{cases}$$

Solving this system yields

$$\alpha = 0.204, \quad \omega_0 = 2.31 \cdot 10^{14}\ \mathrm{s}^{-1}.$$

Given that $\omega_0 = \sqrt{\dfrac{k}{m}}$, we find

$$k = \omega_0^2 m = 1.06 \cdot 10^3\ \mathrm{N/m}.$$

$$\boxed{\alpha k = 216\ \mathrm{N/m}, \quad k = 1.06 \cdot 10^3\ \mathrm{N/m}}$$